Jiajun Fan

CS Ph.D. Student · University of Illinois Urbana-Champaign

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RL Post-Training for Generative Models | Multimodal Reasoning LLMs | Superhuman Deep RL | Agentic RL | Recursive Self-Improvement | Agentic Scientific Discovery

EXPERIENCE

Amazon – AGI Foundations (Audio / Multimodal LLM)

Applied Scientist Intern

Seattle, WA, USA May 2025 – Dec 2025

- **Full-stack distributed RL post-training:** sole owner of the end-to-end pipeline for multimodal reasoning LLMs — training infrastructure, algorithm and reward design, evaluation and model delivery — and sole first author of the resulting publication (CESAR, ICLR 2026).
- Built and scaled a **multi-node, multi-GPU GRPO training stack** for omni-modal LLMs, and designed a process-reward framework that moves RLVR beyond outcome verification and removes any dependence on human-annotated reasoning traces.

EDUCATION

University of Illinois Urbana-Champaign

Ph.D. in Computer Science **GPA 4.0/4.0**

Urbana, IL, USA Aug 2024 – Dec 2027 (expected)

- **Research:** Autonomous RL post-training for flow/diffusion models & multimodal reasoning LLMs
- **Service:** Reviewer @ ICML 2025–26, ICLR 2025–26, NeurIPS 2024–26, COLM 2026, CVPR 2026, WACV 2027, AAAI 2025 & 2027, AISTATS 2025, KDD 2024

Tsinghua University

M.Eng. in Computer Technology **GPA 3.97/4.0 · Top 1.3%**

Beijing, China Sept 2021 – Jun 2024

- **Service:** Reviewer @ NeurIPS 2022–23, ICML 2023–24, ICLR 2024

Nankai University

B.Eng. in Intelligent Science and Technology **Rank 1/83**

Tianjin, China Sept 2017 – Jun 2021

- GPA 93.28/100 (3.9/4.0) · National Scholarship ×2 (Top 1%)

SELECTED PUBLICATIONS * = first / co-first author

COLM 2026

Procedure-Aware Reinforcement Learning for Tool-Augmented Large Language Models

Q. Zheng*, **J. Fan***, C. Cheng, G. Liu

3B model competitive with GPT-5-mini and Claude-Haiku-4.5 on BFCL-V3

TMLR 2026

ProteinZero: Self-Improving Protein Generation via Online Reinforcement Learning

Z. Wang*, **J. Fan***, R. Guo, T. Nguyen, H. Ji, G. Liu

Online RL for self-improving protein design, without human-curated data

ICLR 2026

Incentivizing Consistent, Effective and Scalable Reasoning Capability in Audio LLMs via Reasoning Process Rewards

J. Fan*, R. Ren, J. Li, R. Pandey, P.G. Shivakumar, I. Bulyko, A. Gandhe, G. Liu, Y. Gu

SOTA MMAU · Outperforms Gemini 2.5 Pro & GPT-4o Audio

NEURIPS 2025

Adaptive Divergence Regularized Policy Optimization for Fine-tuning Generative Models

J. Fan*, T. Wei, C. Cheng, Y. Chen, G. Liu

2B SD3 surpasses 4.8B & 12B models

ICLR 2025

Online Reward-Weighted Fine-Tuning of Flow Matching with Wasserstein Regularization

J. Fan*, S. Shen, C. Cheng, Y. Chen, C. Liang, G. Liu

First online RLHF for flow matching models

2026

AudioRSI: Recursively Self-Improving Audio Reasoners

J. Fan*, R. Ren, J. Li, R. Pandey, P.G. Shivakumar, A. Gandhe, Y. Gu, G. Liu, I. Bulyko

The student surpasses its teacher, then becomes it

2026

Training-Free On-Policy Distillation

B. Yang*, **J. Fan***, Z. Zhu, H. Ma, G. Liu

Zero gradient steps · the fused model beats either specialist on average

2025

Fine-tuning Flow Matching Generative Models with Intermediate Feedback

J. Fan*, C. Cheng, S. Shen, X. Zhou, G. Liu · *Under Review*

ICLR 2023 ORAL ★

Learnable Behavior Control: Breaking Atari Human World Records via Sample-Efficient Behavior Selection

J. Fan*, Y. Zhuang, Y. Liu, J. Hao, B. Wang, J. Zhu, H. Wang, S.-T. Xia

Ranked 5/4176 · 24 Atari world records · 100× data efficiency

ICML 2022

Generalized Data Distribution Iteration

J. Fan*, C. Xiao

2× Agent57's mean human-normalized score with 500× less data · 22 world records

NEURIPS 2026

ElegantVLA: Learning When to Think for Efficient Vision-Language-Action Models

Y. Li, H. Liu, K. Ji, Y. Meng, **J. Fan**, Y. Wang, S. Qin, C. Wu, S.-T. Xia, Z. Wang

Up to 2.55× (GR00T) / 3.77× (CogACT) speedup · 13.8 → 26.3 Hz on real-world GR00T tasks

NEURIPS 2026

Variable-Length Generative Protein Design via Generalized Poisson Flow

C. Cheng, Z. Ni, Y. Qu, Y. Chen, R. Guo, **J. Fan**, G. Liu

First on 10 of 16 motif-scaffolding tasks · over 10× more unique successes on several hard targets

ICML 2026

Rethinking the Reranker: Boundary-Aware Evidence Selection for Robust Retrieval-Augmented Generation

J. Sun, P. Jiang, S. Wang, **J. Fan**, H. Wang, S. Ouyang, M. Zhong, Y. Jiao, C. Huang, X. Xu, P. Han, P. Li, J. Huang, G. Liu, H. Ji, J. Han

+10.3% over strong RAG and reranking baselines

ICML 2026

Batched Contextual Reinforcement

B. Yang, H. Ma, **J. Fan**, G. Liu

15.8–62.6% fewer tokens at equal or better accuracy

ICLR 2026

SP-VLA: A Joint Model Scheduling and Token Pruning Approach for VLA Model Acceleration

Y. Li, Y. Meng, Z. Sun, K. Ji, C. Tang, **J. Fan**, X. Ma, S.-T. Xia, Z. Wang, W. Zhu

1.5× lossless speedup (LIBERO) · 2.4× (SimplerEnv)

TPAMI 2026

PRANCE: Joint Token-Optimization and Structural Channel-Pruning for Adaptive ViT Inference

Y. Li, C. Tang, Y. Meng, **J. Fan**, Z. Chai, X. Ma, Z. Wang, W. Zhu

NEURIPS 2025

Variational Supervised Contrastive Learning

Z. Wang, **J. Fan**, T. Nguyen, H. Ji, G. Liu

SOTA among contrastive learning methods · 79.36% Top-1 on ImageNet-1K (ResNet-50)

AWARDS & HONORS

2024 GPA 4.0/4.0, UIUC Ph.D. Program

2024 GPA 3.97/4.0, Top 1.3% — Tsinghua University M.Eng.

2021	Outstanding Graduates — Nankai University Top 1%
2021	Excellent Graduation Thesis — Nankai University
2021	Tang Lixin Scholarship Top 1%
2020	National Scholarship (Rank 1/83) Top 1%
2020	Nomination for Zhou Enlai Scholarship
2019	National Scholarship (Rank 1/83) Top 1%
2019	3rd Prize, RoboCup@HOME Education World Final — Sydney, Australia
2019	Bronze Medal, ACM/ICPC Asia Regional — Xuzhou
2018	National 2nd Prize, National College Students Mathematical Contest in Modeling Top 5%

AT A GLANCE

13+	24	500×	SOTA	592+	12
Top Venue Papers	Atari Records	Data Efficiency vs Agent57	MMAU Audio Reasoning	Scholar Citations	h-index

4.0
UIUC Ph.D. GPA

RESEARCH

Making AI systems that **never stop improving themselves** – progressively removing human scaffolding from RL post-training.

RL Post-Training · Flow/Diffusion RLHF · Audio/Visual Reasoning LLMs · Deep RL (Atari)
Adaptive KL Control · Process Reward Design · Recursive Self-Improvement

REVIEWER

ICML 2023–2024 · **2025 · 2026**

ICLR 2024 · **2025 · 2026**

NeurIPS 2022–2023 · **2024–2026**

COLM 2026 · **CVPR 2026**

WACV 2027

AAAI 2025 · **2027** · **AISTATS** 2025

KDD 2024

OPPORTUNITIES

Seeking research internship – Spring & Summer 2027.
Happy to discuss collaborations on generative model post-training, reasoning LLMs, or RL.